

## Tracing Problems

### Question 1:

```
1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char str1[] = "Hello";
6      char str2[20];
7      char *ptr1 = str1;
8      char *ptr2;
9
10     strcpy(str2, str1); // Copy str1 to str2
11     ptr2 = str2;
12
13     while (*ptr1 != '\0') {
14         *ptr1 = *ptr1 + 1; // Increment each character in str1
15         ptr1++;
16     }
17
18     printf("str1: %s\n", str1);
19     printf("str2: %s\n", str2);
20
21     return 0;
22 }
```

Question 2:

```
1  #include <stdio.h>
2  #include <string.h>
3
4  void reverseString(char *str) {
5      char *start = str;
6      char *end = str + strlen(str) - 1;
7      char temp;
8
9      while (start < end) {
10         temp = *start;
11         *start = *end;
12         *end = temp;
13         start++;
14         end--;
15     }
16 }
17
18 int main() {
19     char str[] = "Pointers";
20     reverseString(str);
21     printf("Reversed string: %s\n", str);
22     return 0;
23 }
```

### Question 3

```
1  #include <stdio.h>
2
3  int compareStrings(char *str1, char *str2) {
4      while (*str1 && *str2) {
5          if (*str1 != *str2) {
6              return 0; // Strings are not equal
7          }
8          str1++;
9          str2++;
10     }
11     return (*str1 == '\0' && *str2 == '\0'); // Check if both strings are at
        their end
12 }
13
14 int main() {
15     char str1[] = "hello";
16     char str2[] = "hello";
17     char str3[] = "world";
18
19     printf("Result 1: %d\n", compareStrings(str1, str2));
20     printf("Result 2: %d\n", compareStrings(str1, str3));
21
22     return 0;
23 }
```

Question 4

```
1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char str[] = "C;Pointers;Strings;Practice";
6      char *token;
7
8      token = strtok(str, ";");
9      while (token != NULL) {
10         printf("%s\n", token);
11         token = strtok(NULL, ";");
12     }
13
14     return 0;
15 }
```